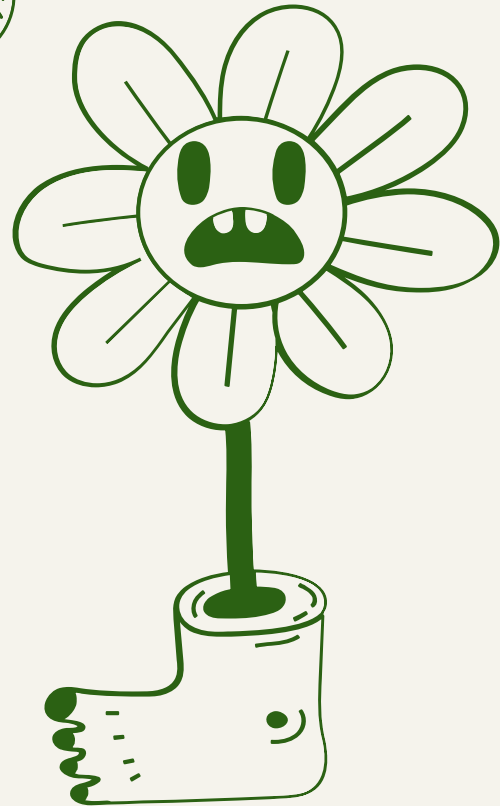
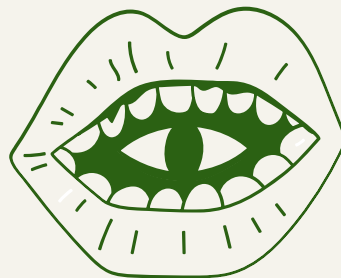


Summarizer

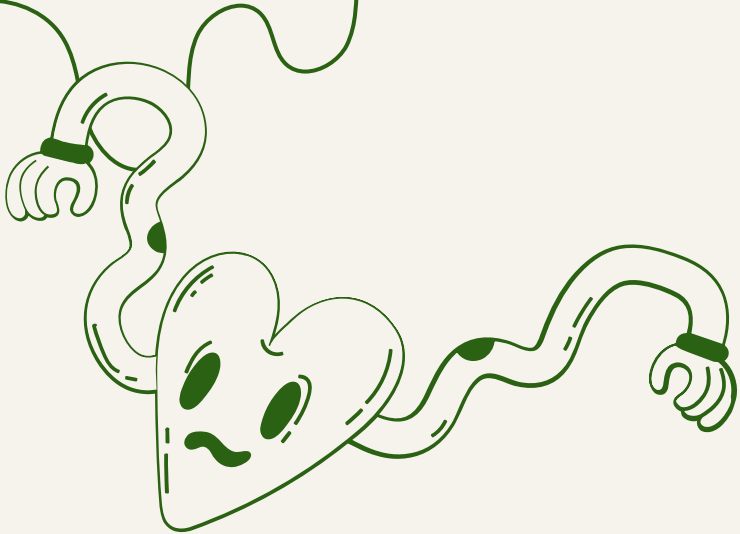
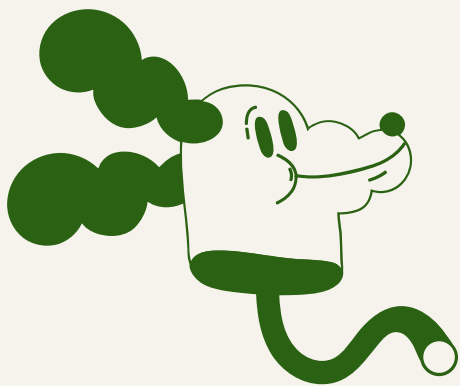
23/03/2023

Tiago Marques

#81 PowRangers

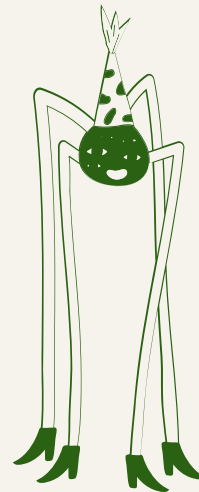
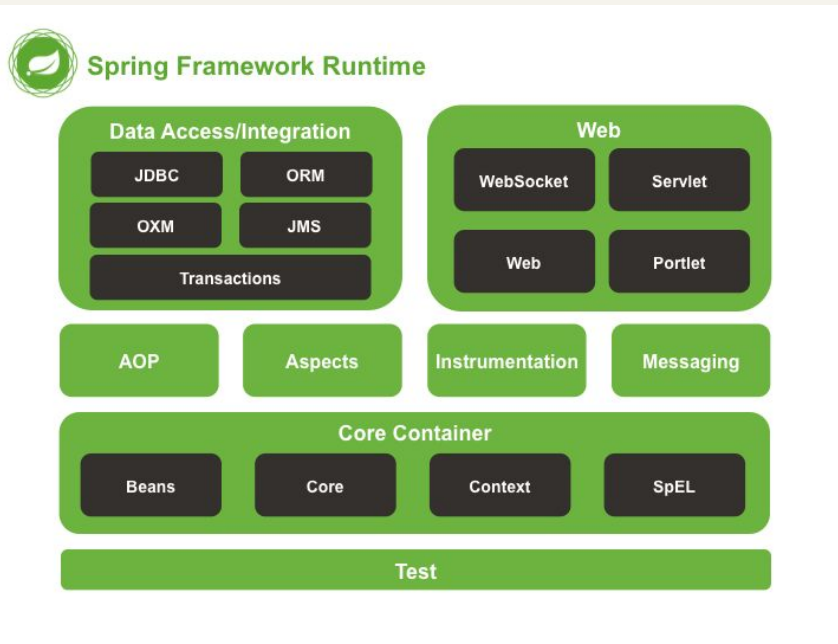


Spring framework



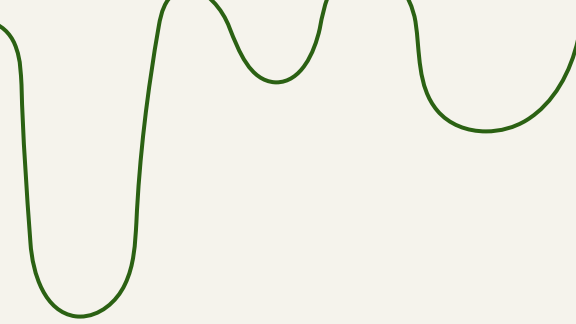
What is the Spring ?

Consists of features organized into about 20 modules, grouped into the Core Container, AOP and Instrumentation, Messaging, Data Access/Integration, Web and Test:



GOOD





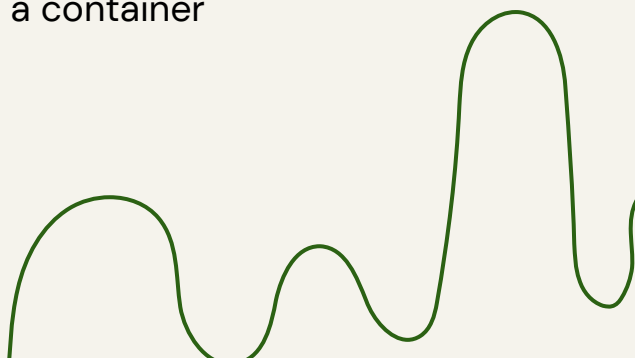
Spring Core

Dependency injection

Dependency injection is a pattern we can use to implement IoC, where the control being inverted is setting an object's dependencies. Connecting objects with other objects, or “injecting” objects into other objects, is done by an assembler rather than by the objects themselves.

Inversion of control

Inversion of Control is a principle in software engineering which transfers the control of objects or portions of a program to a container



Spring Context

Brings everything together,
wiring up the whole
environment.



Spring Beans

Class

Name of the actual implementation class, where is being defined

Conf settings

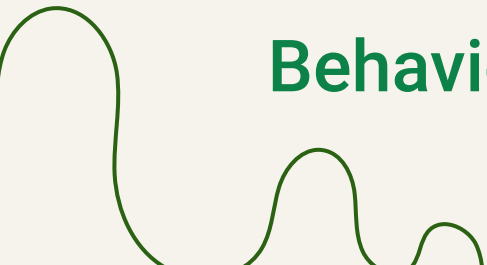
Of newly created object, for ex: number of connection

References

References to other beans

Behavior

Scope - Singleton or prototype
Lifecycle , callbacks, lazy



Metadata Config

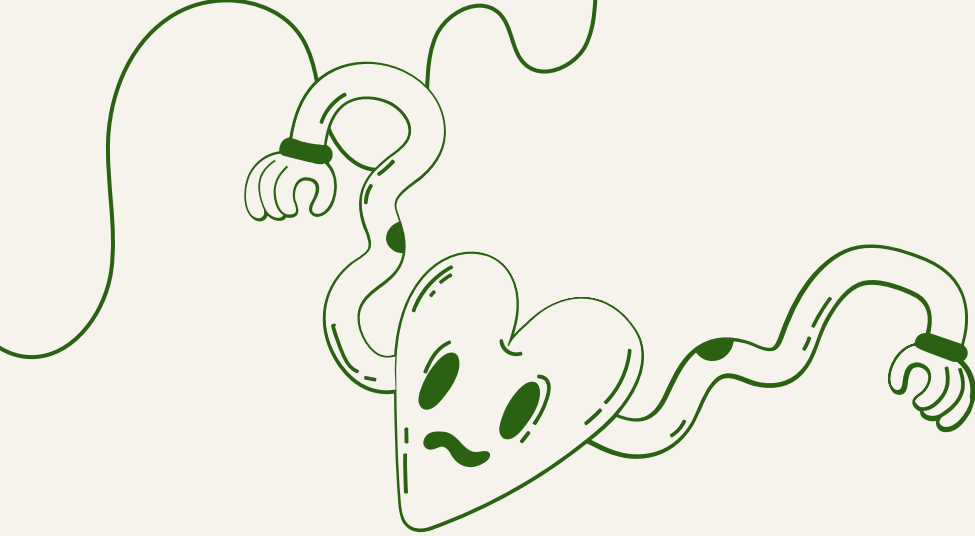
Tells to the container how to instantiate , configure and assemble the objects that compose the application by the following forms;

- **XML -based configuration**

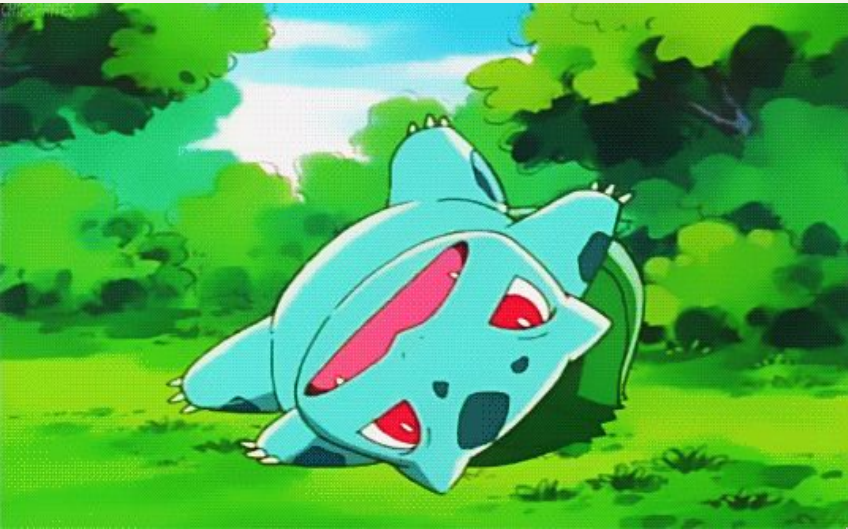
 - Code clean and changes on the fly

- **Annotation -based Configuration**

 - @PersistenceUnit

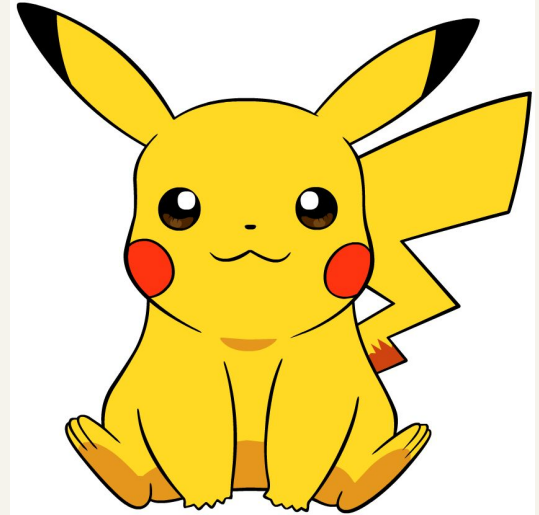


Pokemon Exercise



Pokemon

```
<bean id="p2" class="org.academiadecodigo.bootcamp.model.pokemon.NormalPokemon">
  <property name="nickname" value="Electric Rat"/>
  <property name="type" value="Electric"/>
  <property name="official_name" value="Pikachu"/>
  <property name="moveSet" ref="moveSet2"/>
</bean>
```



Pokemon moves

```
<bean id="moveSet2" class="org.academiadecodigo.bootcamp.model.MoveSet">
  <constructor-arg name="move1" value="ThunderBolt"/>
  <constructor-arg name="move2" value="Thunderwave"/>
  <constructor-arg name="move3" value="Iron Tail"/>
  <constructor-arg name="move4" value="Tackle"/>
</bean>
```



Trainer

```
<bean id="ash" class="org.academiadecodigo.bootcamp.model.Trainer">
  <constructor-arg name="age" value="10"/>
  <constructor-arg name="name" value="Ash"/>
  <constructor-arg name="favPokemon" value="Pikachu"/>
  <property name="pokemons">
    <list>
      <ref bean="p1"/>
      <ref bean="p2"/>
    </list>
  </property>
</bean>
```



Main

```
ApplicationContext context = new FileSystemXmlApplicationContext(  
    "src/main/resources/spring/spring-config.xml");  
TrainerService trainerService = context.getBean(TrainerService.class);  
Trainer trainer = context.getBean("ash", Trainer.class);  
trainerService.add(trainer);  
PokemonService pokemonService = context.getBean(PokemonService.class);
```



Thank You

